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# Natural Language Processing and Computational Linguistics

## Case Study

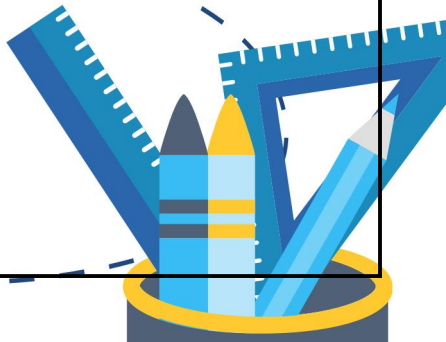

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# Question 2

Explore any NLP lab in any of European Universities, identify their strength, the problem statements, scientists and potential outcomes including research papers from the respective universities.

## Edinburgh NLP Group – University of Edinburgh, UK

# 1. Introduction

Natural Language Processing (NLP) is an area in Artificial Intelligence which allows computers to process human language. NLP has applications in chatbots, translation machines, search engines, voice recognition, summarization techniques, and LLMs (Large Language Models).

Edinburgh NLP Group is one of the recognized research institutions in Europe working in the field of Natural Language Processing. This research institution works in Computational Linguistics as well as in Modern AI. The research areas of Edinburgh NLP Group include language modeling, machine translation, multi-lingual NLP, semantics, parsing, dialogue, information retrieval, speech and multimodal AI. (edinburghnlp.inf.ed.ac.uk)

Edinburgh NLP Group is part of School of Informatics and Institute for Language, Cognition and Computation at the University of Edinburgh.

## 2. Strengths of the Edinburgh NLP Group

The main strengths of Edinburgh NLP are:

**I) Large Language Models**
The scientists explore how LLMs comprehend languages, reason, produce text and deal with long documents. The scientists also research the dependability and limitations of such models.

**II) Machine Translation**
Machine translation is one of the strongest areas of Edinburgh. Researchers from the group created the Nematus neural machine translation system. (edinburghnlp.inf.ed.ac.uk)

**III) Multilingual NLP**
The scientists are interested in developing systems that will work with several languages. It is crucial to consider differences in grammar, vocabulary and amount of available data.

**IV) Multimodal NLP**
Moreover, the scientists study the interactions between language and images, speech and other information. Such research can lead to developing of multimodal AI systems.
**V) Responsible NLP**
The researchers investigate such aspects as bias, misinformation, fairness and dependability to use the NLP systems safely.

## 3. Major Problem Statements

There are several key problem statements in the work of Edinburgh NLP Group.

**Problem 1: Ambiguity of Human Language**
The human language is ambiguous. For instance,
"I saw the man with the telescope"
This phrase has two interpretations. Thus, NLP systems should have an understanding of the context, grammar and semantics.

**Problem 2: Multilingual Translation**
It is quite hard to do translation between languages, particularly for languages with not enough training data.

**Problem 3: Trustworthy Large Language Models**
LLMs can generate wrong information ("hallucinate"). Hence, researchers consider how to make them trustworthy and grounded in facts.

**Problem 4: Multilingual Question Answering**
The question of the user may be asked in one language, whereas the information is best available in another language. Thus, the researchers try to find out how to make retrieval and generation multilingual.

**Problem 5: Responsible AI**
AI models may learn biases from their training data. Hence, researchers investigate the problem of fairness and misinformation.

# 4. Important Scientists

Notable scientists connected with the Edinburgh NLP group are:

**Alexandra Birch**
Alexandra Birch has done research in the field of machine translation and multilingual NLP. Her research focuses on multilingual retrieval-augmented generation. (research.ed.ac.uk)

**Barry Haddow**
Barry Haddow has expertise in machine translation and language technology. He has also researched multilingual and speech technologies. (edinburghnlp.inf.ed.ac.uk)

**Mirella Lapata**
Mirella Lapata is the Professor of NLP. Her research interests include language understanding, language generation, reasoning, and multimodal learning. (research.ed.ac.uk)

**Shay Cohen**
Shay Cohen does research in the fields of language modeling, semantics, syntactic parsing, text generation, and multilingual learning. (edinburghnlp.inf.ed.ac.uk)

**Alex Lascarides**
Alex Lascarides has expertise in discourse, dialogue, language grounding, and human robot interaction. (edinburghnlp.inf.ed.ac.uk)

# 5. Important Research Papers

### 5.1 Multilingual Retrieval-Augmented Generation

A recent study by Leonardo Ranaldi, Barry Haddow and Alexandra Birch, published in 2026, looks into Multilingual Retrieval-Augmented Generation for Knowledge-Intensive Question Answering. In general, RAG retrieves relevant information and then provides it to a language model that generates an answer to the question.

However, the researchers consider the possibility of performing this task in multiple languages. They propose CrossRAG which translates retrieved documents into one common language and then generates a response. The research aims to improve question answering for both high- and low-resource languages. (research.ed.ac.uk)

### 5.2 Dialectic-RAG

Another research work published in 2025 is "Improving Multilingual Retrieval-Augmented Language Models through Dialectic Reasoning Argumentations."

The research is concerned with the issue of contradictory information contained in the retrieved documents. The authors suggest an approach that uses reasoning to compare different evidences before generation. (research.ed.ac.uk)

The above studies show that Edinburgh researchers explore current challenges in the domain of generative AI rather than just traditional NLP.

# 6. Tools and Technologies

Some significant contributions of Edinburgh NLP to NLP tools and technologies include:

I) **Nematus**

Nematus is a neural machine translation system created by the researchers associated with Edinburgh NLP. It has been used for research in the field of neural machine translation. (edinburghnlp.inf.ed.ac.uk)

II) **KenLM**

KenLM is a highly efficient language-modelling toolkit. Language models estimate the probability of word sequences and are applied in many NLP tasks. (edinburghnlp.inf.ed.ac.uk)

Other tools used by researchers include machine learning libraries, multilingual datasets, information retrieval systems and GPU computation.

# 7. Possible Outcomes

The research of Edinburgh NLP has several possible practical outcomes.

**1. Improved translation:**
Enhanced multilingual NLP can provide access to information for speakers of different languages.

**2. Reliable AI:**
RAG and evidence-based reasoning can assist in minimizing wrong responses from language models.

**3. Multilingual AI assistants:**
Users can ask questions in their language, and the system would find information from sources in other languages.

**4. Healthcare applications:**
NLP is capable of analysing medical information and providing useful information for healthcare specialists.

**5. Social media analysis:**
NLP can be used to identify misinformation, abusive language and harmful content.

**6. Human-robot interaction:**
Language understanding can enable robots to comprehend human orders.

## 8. Relevance to India

The research conducted at Edinburgh NLP is extremely relevant to India since there are a large number of languages in India.
An Indian user can pose a query in Tamil, whereas important information can be found in English or Hindi. The multilingual RAG model will retrieve information in various languages and deliver the response in Tamil.

The following are some areas where this kind of technology can prove useful:

- Education
- Healthcare
- Government services
- Banking
- Agriculture
- Customer support

This means that research in multilingual NLP is extremely relevant to Indian regional languages.

## 9. Conclusion

Edinburgh NLP Group at the University of Edinburgh is a leading European research group that works on Natural Language Processing and Computational Linguistics. The main areas of its expertise are machine translation, multilingual NLP, Large Language Models, language understanding, multimodal AI, and responsible NLP.

Scientists like Alexandra Birch, Barry Haddow, Mirella Lapata, Shay Cohen and Alex Lascarides investigate different issues related to NLP. Their research concerns such problems as language ambiguity, multilingual translation, LLMs' reliability, information retrieval and responsible AI.

Recent research on Multilingual Retrieval-Augmented Generation and Dialectic-RAG exemplifies how NLP research adapts to the era of generative AI.
All in all, the research done by Edinburgh NLP illustrates how university research contributes to development of AI applications. Its applications can be used for translation, education, healthcare, multilingual assistants, social-media analysis and human-robot interaction. The research conducted by this university is extremely relevant to India, where multilingual AI can serve as a bridge between technology and users speaking regional languages.

# 10. References

1) Edinburgh NLP Group, University of Edinburgh – Official Research Group. Edinburgh NLP

2) Ranaldi, L., Haddow, B., & Birch, A. (2026). Multilingual Retrieval-Augmented Generation for Knowledge-Intensive Question Answering Task. Findings of EACL 2026.University of Edinburgh Research Explorer

3) Ranaldi, L., Ranaldi, F., Zanzotto, F. M., Haddow, B., & Birch, A. (2025). Improving Multilingual Retrieval-Augmented Language Models through Dialectic Reasoning Argumentations. EMNLP 2025.
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4) University of Edinburgh – Mirella Lapata Research Profile.
Mirella Lapata – University of Edinburgh.